

CLAUDE.md — SCX Project

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Project Overview

SCX (State-Conditioned eXpertise) is a mathematical theory + engineering framework. Its core discovery is: **distinguishing label noise from sample intrinsic difficulty — this problem has been mathematically proven unsolvable without assumptions.** Under minimal sufficient assumptions, multi-expert consistency can provide exact constant minimax optimal noise detection.

Paper Matrix (All Targeting Nature Series)

#	Directory	Journal	Status
I	paper/nature_theory	Nature	Main text drafted, SI Awaiting translation
II	paper/nature_curation	Nat Comp Sci	Concept designed
III	paper/paper1_nature	Nat Mach Intell	Draft available, awaiting GPU
IV	paper/paper2_mlip/	npj Comp Mat	Completed, awaiting supercomputing

See paper/PAPER_MATRIX.md.

Core Theorems

#	Content	File	Verification Status
1	Noise Detection Guarantee (F1 exponential convergence)	theory/theorems/01_noise51detection_guarantee.md	
2	Weak Feature Failure Boundary ($\delta \rightarrow 0$ degradation)	theory/theorems/02_weak_51feature_failure.md	
3	Noise-Difficulty Unidentifiability	theory/theorems/03_unide51ntifiability_theorem.md	
4'	Exact Constant Minimax Optimality	theory/explorations/exact51constant_minimax.md + 4 lemma files	
5	Cluster Consistency (k-means state discovery)	theory/explorations/clust51er_consistency_v3.md	
6	Bootstrap Stability Diagnostics	theory/explorations/feat51ure_strength_via_stabilit	

Authoritative reference: `theory/THEOREMS_UNIFIED.md`.

Critical Path

paper/nature_theory/	← Primary: mathematical theory, Nature submission
paper/nature_curation/	← Secondary: practice principles, Nat Comp Sci
paper/paper1_nature/	← Tertiary: method validation, Nat Mach Intell
scx-life/	← Application modules (health/ + drug/)

Current Blockers

- GPU unavailable → Paper III AIN v3 retraining, CIFAR/MedMNIST full training
- Nature SI needs LaTeX translation ($\sim 12,000$ lines md $\rightarrow$ tex, ~ 86 pages)

Connections

- HPC management: `../.hpc/hpcmgr.py` (job submission/monitoring)

- HPC drives: W: $\rightarrow$ cxshu, X:/Y:/Z: $\rightarrow$ xiyang
- Knowledge base: **knowledge/** (Obsidian vault)